Definite integrals

1

a $e^2 - e^{-4}$

b $\frac{e^{12}}{4} - \frac{1}{4}$

c $\frac{2e^3}{3} - \frac{2}{3}$

d $\frac{1}{3}e^{12} - \frac{1}{3}e^{-6}$

e $-\frac{1}{2}e^{-10} + \frac{1}{2}e^6$

f $13 + e^2$

g $\frac{2e^{15}}{5} - \frac{e^{-21}}{7} - \frac{9}{35}$

h $4e^{\frac{3}{4}} + 15e^{-\frac{3}{5}} - 4e^{-\frac{3}{4}} - 15e^{\frac{3}{4}}$

i $\frac{1}{4}e^{16} + 4e^8 - \frac{1}{4}e^{12} - 4e^6 + 16$

j $\frac{e^8}{2} + \frac{2e^{12}}{3} + \frac{e^{16}}{4} - \frac{e^{-8}}{2} - \frac{2e^{-12}}{3} - \frac{e^{-16}}{4}$

k $-\frac{1}{3}e^{-9} + \frac{1}{2}e^6 + \frac{1}{3}e^{12} - \frac{1}{2}e^{-8}$

l $\frac{1}{4}e^{64} - \frac{1}{4}e^4 + 1323$

2

a 1

b -1

c $2\sqrt{3}$

d -2

e 0

f 0

g $2\sqrt{3}$

h $2\sqrt{3} + 2$

i $\frac{6}{\pi}$

j 0

k $\frac{1}{2} - \frac{1}{\sqrt{2}}$

l 0

m $-\frac{\sqrt{3}}{2} + \frac{3}{2}$

n $\frac{\sqrt{3}}{4} + 4$

o -6

p $\frac{3}{16}$

q 0

3

a $y' = e^x - e^{-x}$

b $\ln\left(e^3 + e^{-3}\right) - \ln 2$

4

a $y' = 3xe^{3x}$

b $\frac{26e^{27} - 17e^{18}}{9}$

Graph symmetry

5

a $\left(\frac{e^8}{4} - \frac{e^{-4}}{4}\right)$ units^2

b $(e^2 - 1)$ units^2

c $(e^5 - 1)$ units^2

d $(e^4 + 15)$ units^2

e $\left(\frac{e^{12}}{6} - \frac{e^{-6}}{6}\right)$ units^2

f $(2e^9 - 2e^{-9})$ units^2

g $(4e^{\frac{1}{8}} - 4e^{-\frac{1}{8}})$ units^2

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6

a 6 units^2 **b** $\frac{2}{3} \text{ unit}^2$ **c** 1 unit^2 **d** 12 units^2

e 1 unit^2 **f** 3 units^2 **g** 4 units^2 **h** $6\pi \text{ units}^2$

i $\left(\frac{\pi}{9} + 3\right) \text{ units}^2$ **j** $\left(\pi + \frac{1}{4}\right) \text{ units}^2$

k $(\sqrt{2} + 2) \text{ units}^2$

7

a $x = 0$ **b** $f(x) \to e^{3x}$

c Odd **d** $\left(\frac{1}{3}e^{9} + \frac{1}{3}e^{-9} - \frac{2}{3}\right) \text{ units}^2$

e $\left(\frac{2}{3}e^{3} + \frac{2}{3}e^{-3} - \frac{4}{3}\right) \text{ units}^2$

8

a 0 **b** **i** $0 \leq x \leq \frac{\pi}{12}$ **ii** $-\frac{\pi}{12} \leq x \leq 0$

c $\frac{1}{4} \text{ unit}^2$

9

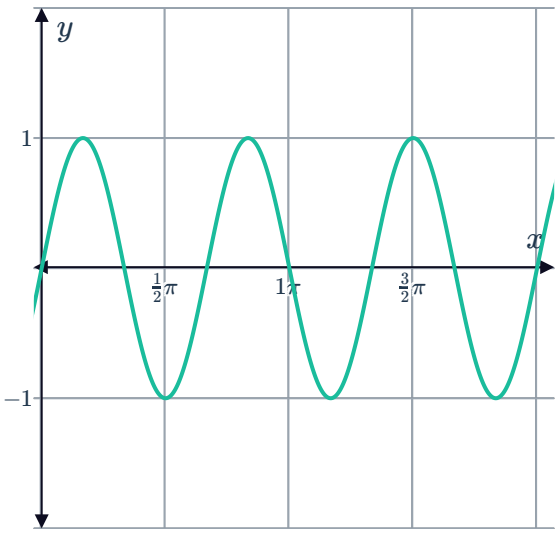
a $2\sqrt{2 - \sqrt{2}} \text{ units}^2$ **b** 16 units^2

10

a 10 radians **b** $\frac{10\sqrt{2}}{\pi} \text{ units}^2$

11

a

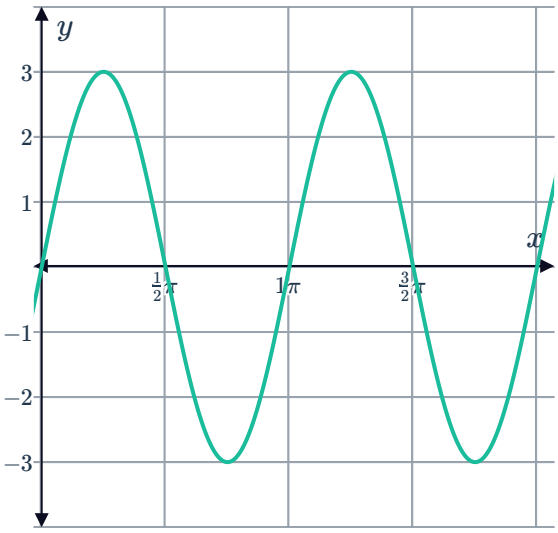


b $\left(\frac{\sqrt{2}}{6} + \frac{1}{3}\right) \text{ units}^2$

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12

a



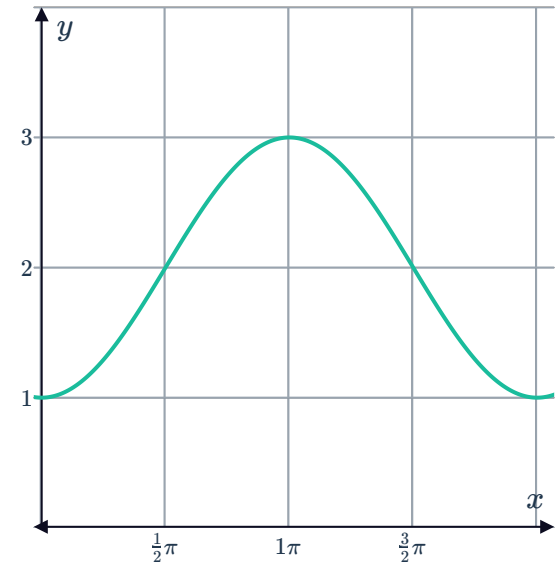
b

i 3 units^2

ii $\frac{9}{4} \text{ units}^2$

13

a



b $(\pi - 1) \text{ units}^2$

Use technology

14 20.61

15 0.65 units^2

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